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A palaeomagnetic study of the Upper Mesozoic succession in Northern Tunisia

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Summary. The Upper Mesozoic section from Northern Tunisia provided an Upper Jurassic palaeomagnetic pole of 65.2°S 20.3°E $\alpha_{95} = 6.1$ calculated from the means of normal and reversely magnetized samples from the uppermost Callovian, Oxfordian, Kimmeridgian and Portlandian rocks. In general the only Cretaceous rocks to yield acceptable results were the few samples collected from fresh outcrops.

A polarity sequence can be established for the Upper Jurassic which can be correlated with the oceanic Keathley anomaly sequence. One consequence of the proposed correlation of the oceanic anomaly with the terrestrial palaeomagnetic sequence is to suggest a slightly different age for the Oxfordian-Kimmeridgian boundary. One interpretation of the frequent intermediate directions of magnetization in the Cretaceous sequence is that there may be a number of unrecognized short period reversals within the Cretaceous and, more particularly, during the so-called Cretaceous normal period.

Introduction

In this paper we report the results of a magnetostratigraphic investigation of the Upper Mesozoic section in Tunisia. Samples were collected from two principal locations southwest of Tunis (Fig. 1) from rocks which ranged in age from Upper Callovian to Maestrichtian. The Upper Cretaceous samples were collected on the slopes of Djebel Ben Salah beyond the limits of the town of Pont du Fahs, in a section which has been proposed as a hypostratotype (Salaj 1974). The Upper Jurassic was sampled in a recent shallow road-cut at the western end of Djebel Oust and the Lower Cretaceous from a stream section east of the djebel.

As a preliminary to the study a review of the gross behaviour of the geomagnetic field during the late Mesozoic was carried out utilizing published radiometric ages assigned to rocks for which polarity data were available. This we believe a more satisfactory method than the syntheses of all data of the type carried out by McElhinny & Burek (1971) and

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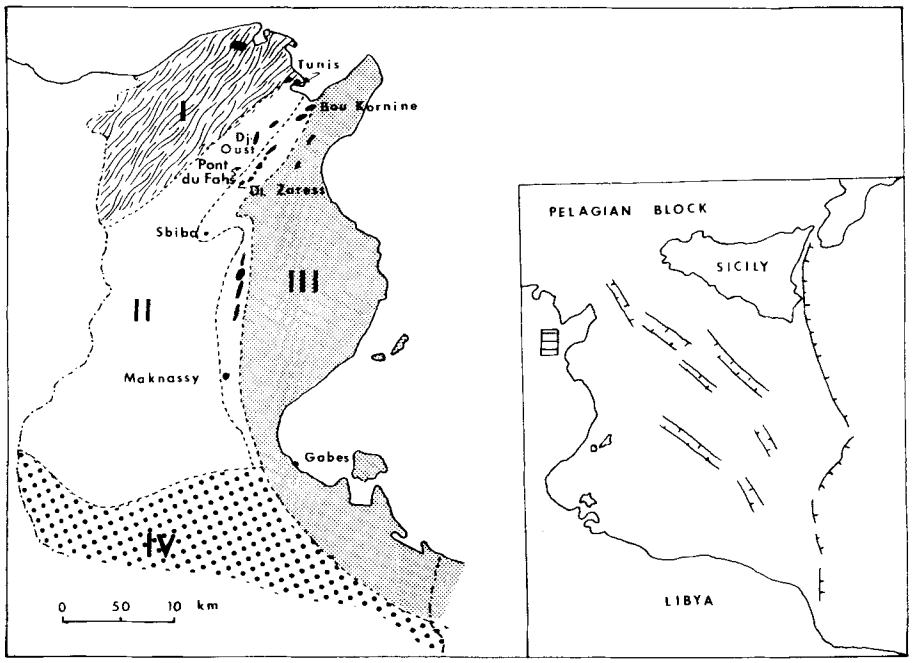


Figure 1. Structural regions of Northern Tunisia simplified from Burollet *et al.* (1978). (I) The northern Tunisian Alpine zone. (II) Zone of the Tunisian Atlas and intermediate chains. (III) The Eastern Platform bordered to the west by the zone of the north-south axis. (IV) The Saharan platform. Samples examined were collected in the zone marked by horizontal lines.

Creer (1971). It differs slightly from the earlier review of Irving & Pullaiah (1976) in as much as all polarity information from biostratigraphically dated sequences was excluded. The generalized polarity history was found by calculating a running mean with a 5 Myr window for 1.66 Myr intervals and by imposing a simple weighting procedure (normal polarity 1, reversed 3, mixed 2). The nature of the available data, from which it is not always clear whether a given date represents a time of intrusion or extrusion of a few or of a series of units, and where polarity is mixed, how many reversals are recorded, precludes any more sophisticated technique.

If radiometric data were randomly distributed then with a 5 Myr window a running average of two would indicate a reversal frequence close to that of the Tertiary (Cox 1968), yet the actual average is significantly different from this (Fig. 2). There is an evident long normal interval extending from 77 to 112 Myr the limits of which are not very different from the 81 to 110 Myr resulting from the Irving & Pullaiah analysis (1976) or from Helsley & Steiner (1969). Below there is found a zone of mixed polarity extending back to at least 152 Myr and which has been called the Serra Geral interval by Creer (1962).

The palaeomagnetic study was based upon several hundred samples collected in northern Tunisia during the 1974 and 1975 field seasons. The majority of cores were drilled in the field with a gasoline powered drill but block samples were collected for subsequent coring in the laboratory on the occasion when the drill failed. All orientation was carried out with a Brunton compass. With the exception of a single lava flow, all the samples consisted of limestone or dolomite. Cores were sliced to standard sized cylinders with replicate cores where possible.

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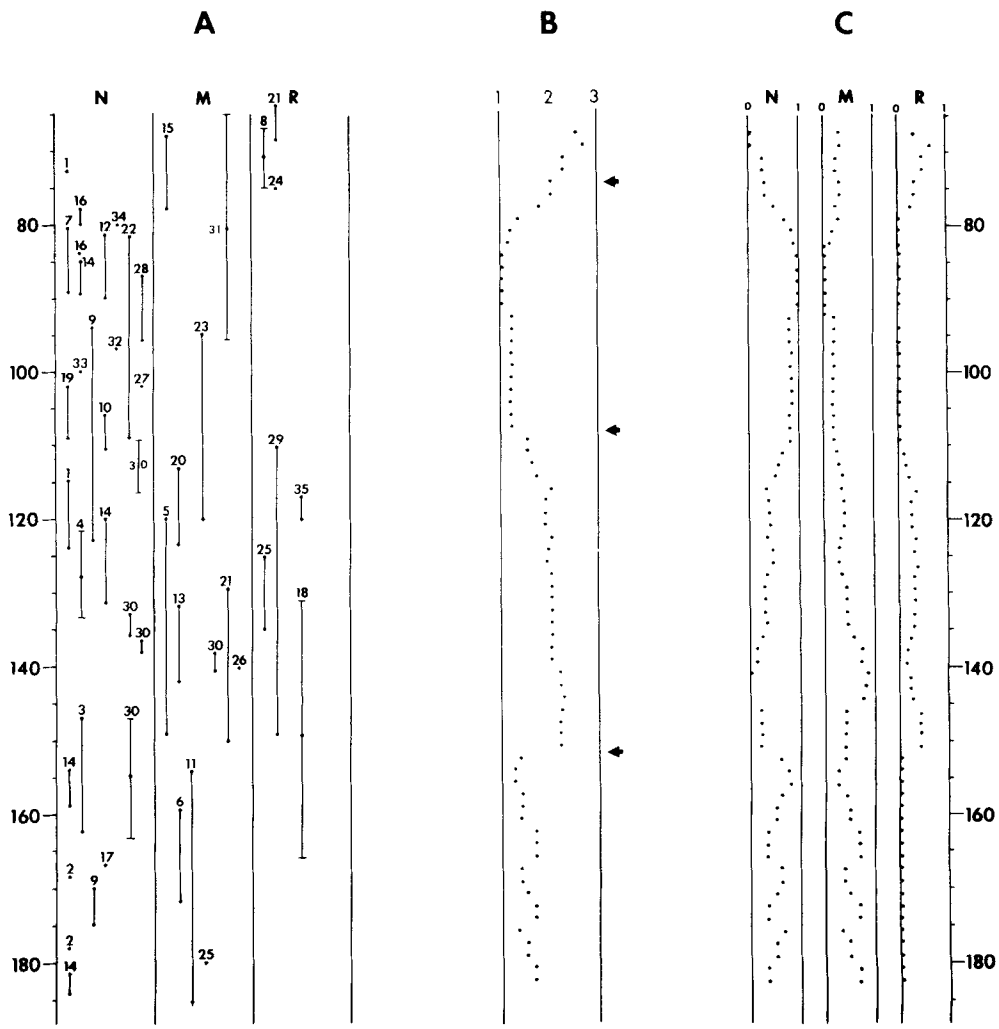


Figure 2. Polarity synthesis of radiometrically dated igneous rocks. (A) Age range and polarity (N = normal, M = mixed, R = reversed). (B) Running average of polarity (see text) (N = 1, M = 2, R = 3). Arrows indicate the limits of the normal interval and the underlying Serra Geral mixed polarity interval. (C) Fraction of normal, mixed and reversed sites. The numbers in A refer to the data sources as follows: (1) Beck, Ellis & Beske (1972). (2) Boesen, Irving & Robertson (1961). (3) Bull, Irving & Willis (1962). (4) Briden (1967). (5) Creer (1962); Creer *et al.* (1965); McDougall & Ruegg (1966). (6) Creer, Mitchell & Abou Deeb (1972). (7) Currie, Grommé & Verhoogen (1963). (8) Deutsch & Kristjansson (1974). (9) Gidskehaug, Creer & Mitchell (1975). (10) Gough & Opdyke (1963). (11) Graham & Hales (1957). (12) Grommé & Merrill (1965). (13) Grommé, Merrill & Verhoogen (1967). (14) Hailwood & Mitchell (1971). (15) Hanna (1973). (16) Harrison *et al.* (1965). (17) Irving (1963). (18) Krumsiek, Nagel & Nairn (1968). (19) Larochelle, Black & Wanless (1965). (20) Larochelle (1968). (21) Linares & Valencio (1975). (22) McDougall & McElhinney (1970). (23) MacDonald & Opdyke (1972). (24) Opdyke & MacDonald (1973). (25) Opdyke & Wensink (1966). (26) Robertson (1963). (27) Robertson & Hastie (1962). (28) Scharon & Hsu (1969); Zartman *et al.* (1967). (29) Spall (1968). (30) Symons (1973). (31) Symons (1974). (32) Symons (1973b). (33) Teissere & Beck (1973). (34) Van der Voo (1969). (35) Valencio (1972). Error bars represent the published precision limits of the published ages.

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An approximate calculation based upon published rates of carbonate sedimentation (Newell 1955; Cloud 1959) and the Tertiary polarity record suggests that during one of the shorter polarity intervals the thickness of limestone deposited may be a range between 25 and 45 cm. As the greater part of the exposed sequence consisted of thin limestone and shale alternations about 10 cm thick we chose to sample every limestone band. There was thus dense sampling where the limestone and shale alternated rapidly, but no representation of those intervals where only shale was found in outcrop.

All remanence measurements were performed with a Digico complete results magnetometer. All samples were subjected to magnetic cleaning either by using a Schonstedt GSD 1 or a furnace housed in a doubly shielded mu metal room in which the residual field was reduced to below 20γ by ancilliary coils.

General geology

The eastern Tunisian platform, the western part of the Pelagian block (Burollet 1967; Burollet, Mugniot & Sweeney 1978), began to differentiate from the subsiding zone of the middle and southern Atlas in mid-Jurassic time. Salaj (1978) dates the first movements as Upper Sinemurian and from detailed sedimentological studies Ehrlich & Cossey (1978) show that in the region of Djebel Bou Kornine movements which began in Bathonian times died out in the Oxfordian. The zone where this differentiation is most clearly distinguishable is referred to as the north–south axis by Burollet (1967) and later authors. Along this axis a number of regional highs supported Tithonian reefs and these are shown in black in Fig. 1.

One evidence of this downward movement is that the ammonitico rosso, commonly regarded as a deep-water deposit, rests directly upon older shallow water limestone facies. The vertical displacement between Bathonian and Upper Callovian-Oxfordian along the Bou Kornine faults is nearly 400 m (Ehrlich & Cossey 1978), and other fault systems of comparable displacement and ages may be presumed to have existed. The platform to the east remained a relatively positive area as is evidenced by the occurrence of condensed, shallow water sequences. To the west within the subsiding basin deposition was continuous and a relatively thick succession is found. The sections from which material was derived for this study, that is Djebel Oust and Djebel Ben Salah, lay close to the eastern margin of this subsiding basin. The sections are tectonically undisturbed with dips seldom exceeding a few degrees.

The zonation of both the Upper Jurassic and the Cretaceous is based primarily upon microfaunal studies (Bonnefous 1967; Salaj 1976), and we are grateful to Dr J. Salaj, then with the Geological Survey of Tunisia, for locating these boundaries in the field. The Cretaceous samples from the hypostratotype section (Salaj 1974) are deposited with the geological survey and key samples can be correlated directly with microfaunal collecting sites which provide age control.

To the south the region of the central and southern Atlas and the Tunisian platform gives way to the stable African or Saharan craton of Tripolitania and extreme southern Tunisia. Formational names have been assigned to the Upper Mesozoic lithologies but there seems little general accord in their usage, and they are not always readily identified. In this study we have therefore followed the example of Salaj (1978) avoiding the use of formation names by the use of the standard stage terms.

Results

It is convenient to discuss the study of the Upper Mesozoic in three parts, the Upper Callovian–Oxfordian ammonitico rosso, the Kimmeridgian–Portlandian and Berrisian section and the

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remainder of the Cretaceous section. Lithologically the Upper Callovian–Oxfordian section with its condensed, nodular red limestone sequence is quite distinct from the remainder of the limestones sampled and, while data from the Malmian rocks respond to magnetic cleaning, the Cretaceous rocks on the whole do not.

THE AMMONITICO ROSSO FACIES

In the Mediterranean region the term ammonitico rosso is applied to a red nodular limestone lithology. Manganese nodules and coatings are also found within the ammonitico rosso, which derives its name from the abundance of red stained ammonites found within it. It is a facies which Jenkyns & Torrens (1971) regard as one typical of sea mounts, although Salaj (private communication) has pointed out that several varieties of limestone are grouped under the term.

There are two principal horizons at which the ammonitico rosso is known to occur, one in the Liassic, and the one with which we are concerned in the Malm. According to Spath (1913) the Tunisian ammonitico rosso, as in western Sicily, is Callovian–Oxfordian in age and, although this was questioned by Pervinquière (1903), Castany (1951) and Bonnefous (1972), it was confirmed by the discovery in the basal nodular band of the Upper Callovian *Reineckia* (*Kellawaysites*) – cf. *multicostata* (Peticlerc) – in the Poste Optique section near Zaghouan and *Reineckia* (*Kellawaysites*) sp. and *Oppelia* (*Oxycerites*) sp. in the basal limestone in the Djebel Zaress section (Biely & Rakus 1969).

In Tunisia the typical red nodular limestones are interbedded with red fissile shale which often forms an appreciable part of the section. For this reason no samples were collected from the thickest section opposite hill 422 near Zaghouan. The Poste Optique section was also thin with relatively few limestone bands so that most of the samples were obtained from Djebel Oust road cut where about 23 m of section is exposed and from an approximately equal thickness cropping out on the slopes of Djebel Zaress (Mencik, Salaj & Stranik 1970). The two locations north-east and south of Pont du Fahs are approximately 60 km apart. Every limestone horizon at both localities was sampled, but the shales were too fissile to yield coherent samples. In the thicker bands in the Djebel Oust section closely spaced (2–4 cm) samples were collected.

The ammonitico rosso limestone responded both to alternating field and thermal demagnetization, with dispersion minima at 250 Oe and at 300°C. Numerous pilot samples were progressively demagnetized at 50 Oe intervals to 250 Oe, then at 350, 500, 750 and 1000 Oe or at 100°C intervals up to 650°C or until the core disintegrated. The remaining samples were then demagnetized in three steps bracketing the optimum temperature or demagnetizing field.

The stereographic projection of the remanence directions before and after cleaning are shown in Fig. 3, and the same directions in stratigraphic order illustrated in Fig. 4. In the latter figure different symbols are used to identify samples cleaned by alternating field techniques (·) from those thermally demagnetized (*). Measurements of intensity frequently showed quite rapid changes, so that smooth demagnetization curves are not the rule (see Fig. 5), however, it proved possible to draw stable end point curves (e.g. Fig. 6) for some samples.

THE KIMMERIDGIAN TO LOWER VALANGINIAN SECTION

The rocks in this succession follow the ammonitico rosso without break in the Djebel Oust road section. The section is a fairly uniform alternation of shale and limestone in bands of

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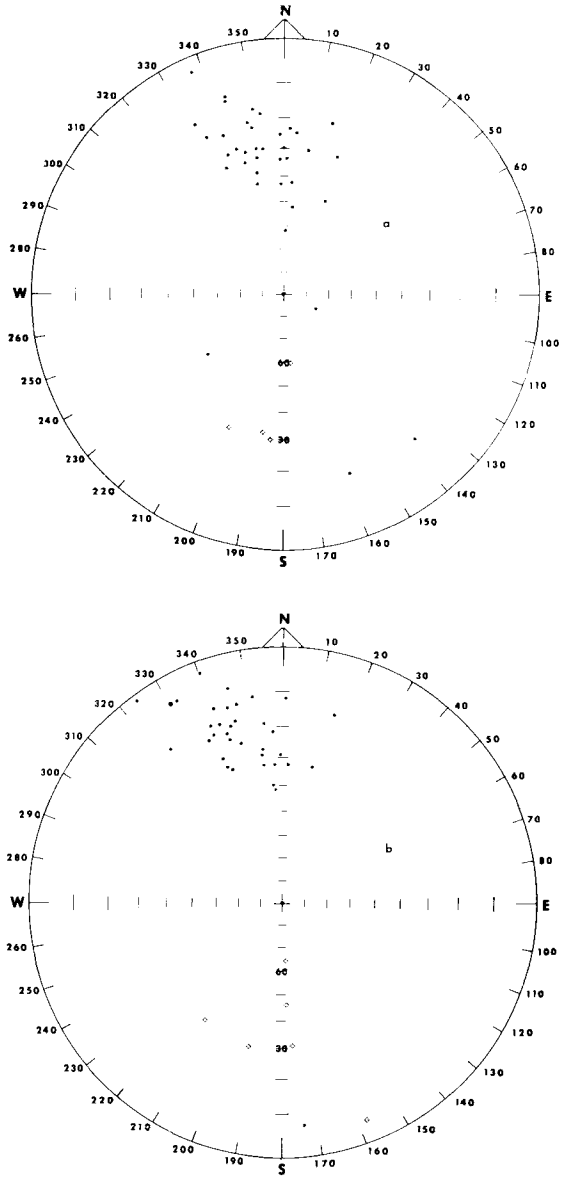


Figure 3. Stereograms of the declination and inclination of samples from the Upper Callovian–Oxfordian ammonitico rosso Djebel Oust. (a) NRM, (b) cleaned directions.

approximately equal thickness. Only the middle Kimmeridgian is marked by a thick uninterrupted shale sequence capped by a limestone 2–3 m in thickness. The limestone bands are most often between 10 and 20 cm in thickness and can sometimes be traced laterally into a band of nodules before disappearing. Every limestone band was cored, and in the thicker bands several cores were obtained.

The Kimmeridgian, Portlandian and Berriasian samples were subjected both to thermal and alternating field cleaning techniques. The least dispersion was found following demagnetization at 350°C or after 350 Oe. The NRM intensity of most samples ranged between 1

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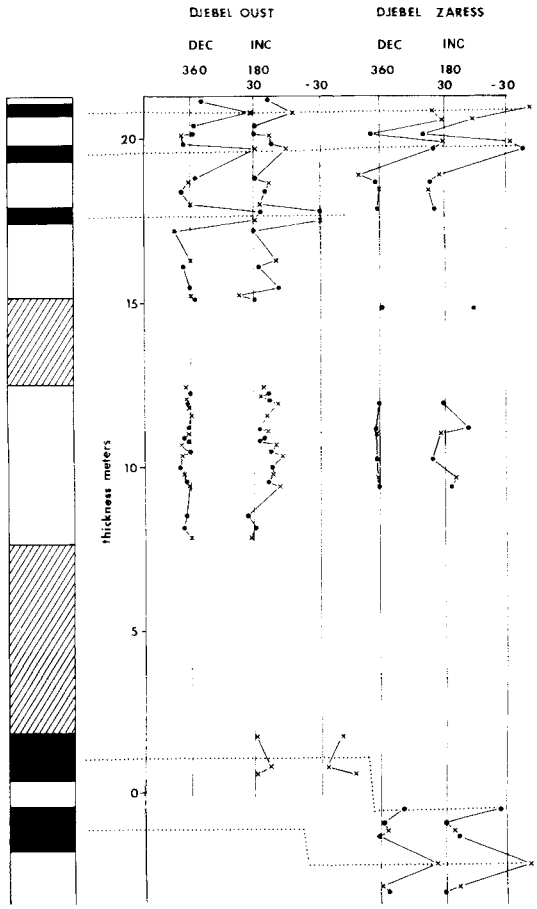


Figure 4. Stratigraphic summary of the declination and inclination values of ammonitico rosso samples from Djebel Zaress near Pont du Fahs and from Djebel Oust, with a polarity synthesis. Obliquely lined areas indicate regions not sampled (zones of friable shale).

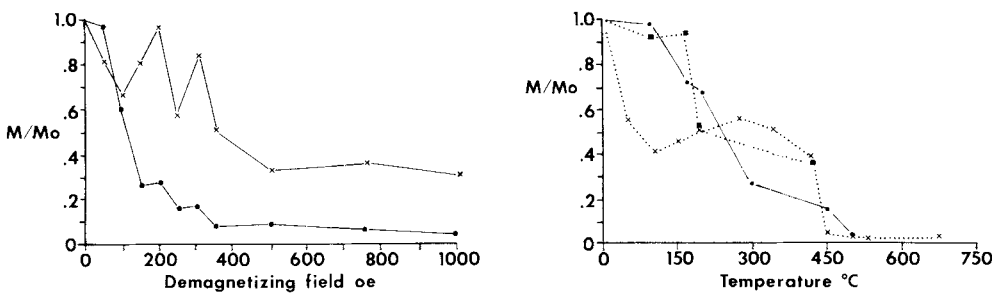


Figure 5. Normalized demagnetization curves of some ammonitico rosso samples.

and 10×10^{-6} emu cm^{-3} but on occasion samples with twice those intensities were found. The intensity of magnetization after cleaning was even more variable, ranging from as low as 0.04 to 4×10^{-6} emu cm^{-3} with the most common values in the middle of the range. While the intensity of magnetization of the cleaned samples was on the average an order of

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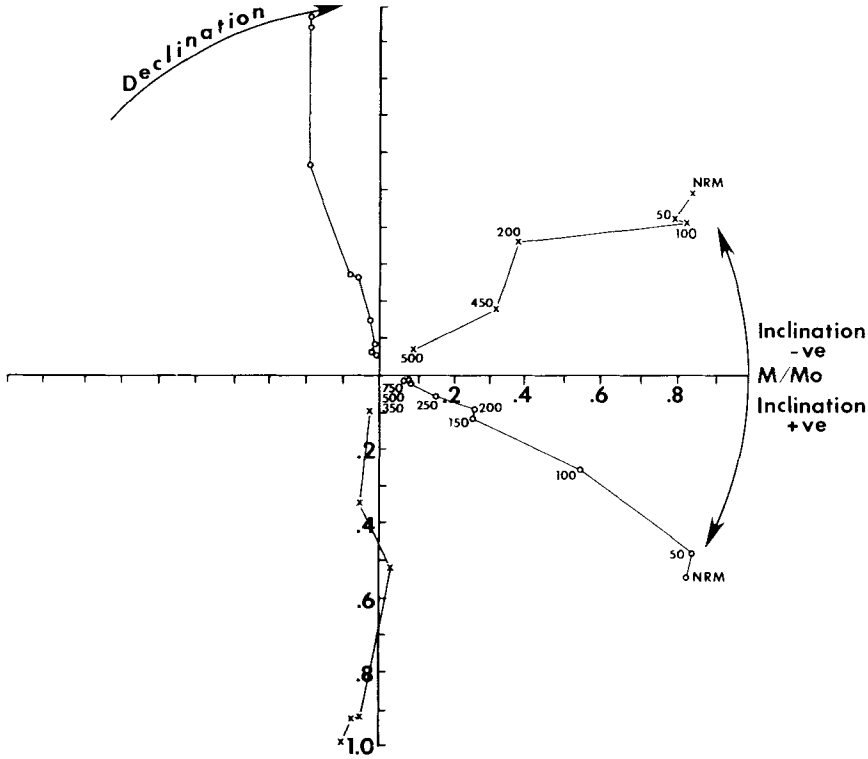


Figure 6. Stable end point curves of a normally magnetized (○) and reversely magnetized (×) samples of ammonitico rosso demagnetized by alternating field and thermal techniques respectively.

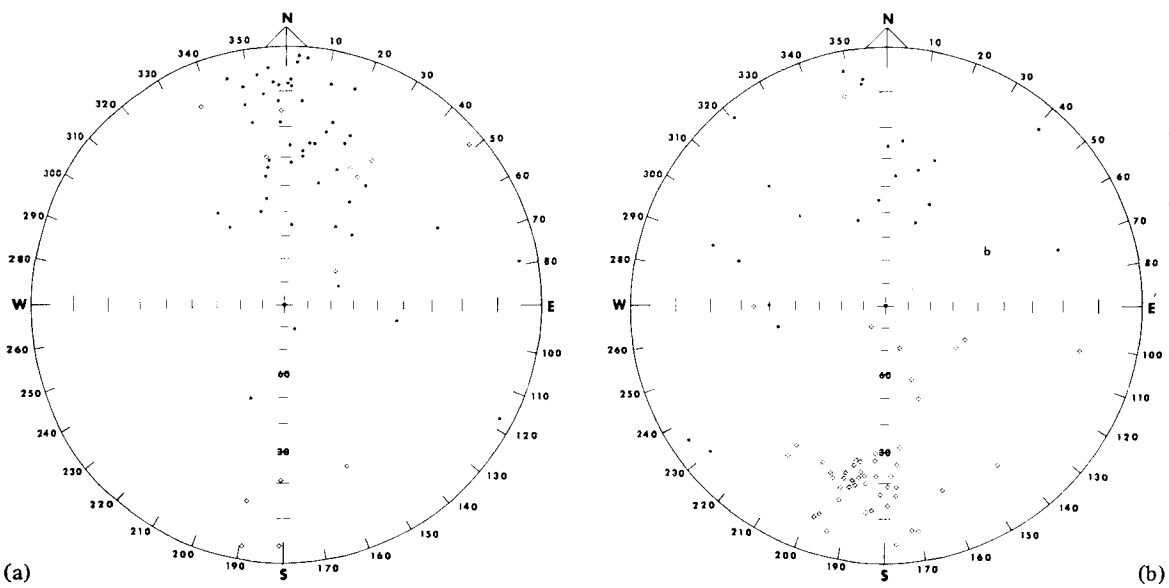
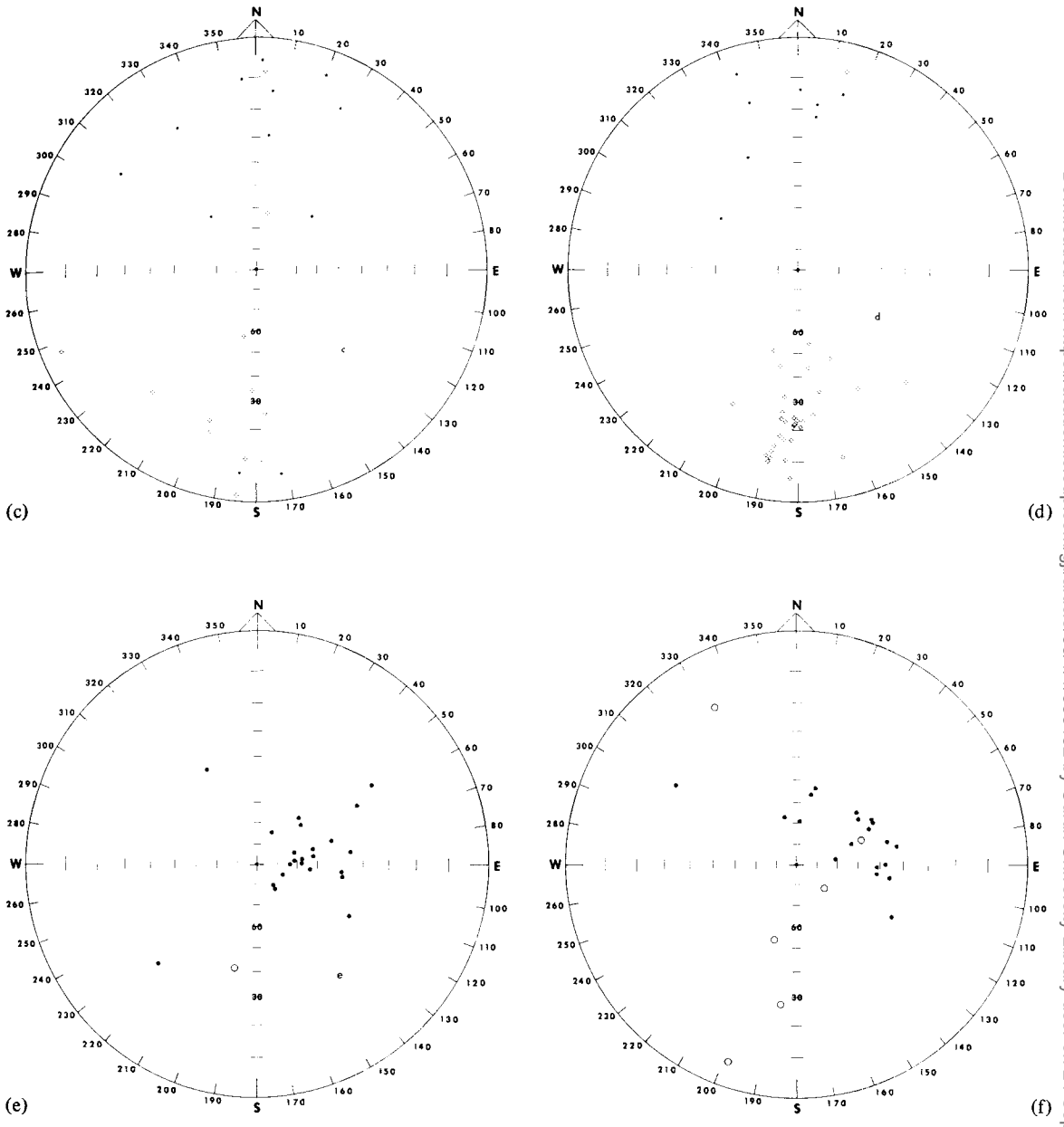


Figure 7. Stereograms of the declination and inclination of Kimmeridgian and Portlandian and Berriasian samples: (a) NRM Kimmeridgian samples*; (b) cleaned Kimmeridgian samples; (c) NRM Portlandian samples*; (d) cleaned Portlandian samples; (e) NRM Berriasian samples; (f) cleaned Berriasian samples. *NRM directions not shown for all samples illustrated in the cleaned directions figures.

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magnitude lower than the NRM intensity, progressive demagnetization, particularly in alternating fields, seldom showed a steady decline in magnetic intensity. Stereograms of the cleaned directions of magnetization and the NRM directions are illustrated in Fig. 7. The directions of magnetization are stratigraphically ordered in Fig. 8, with oblique ruling indicating the thicker shale zones barren of limestone. The Upper Kimmeridgian and Portlandian are characterized by reversed polarity; the Berriasian by normal directions of magnetization. The mean directions of normal and reversed magnetization have been calculated by stage and, as can be seen from Table 1, the direction are close to being antipodal.

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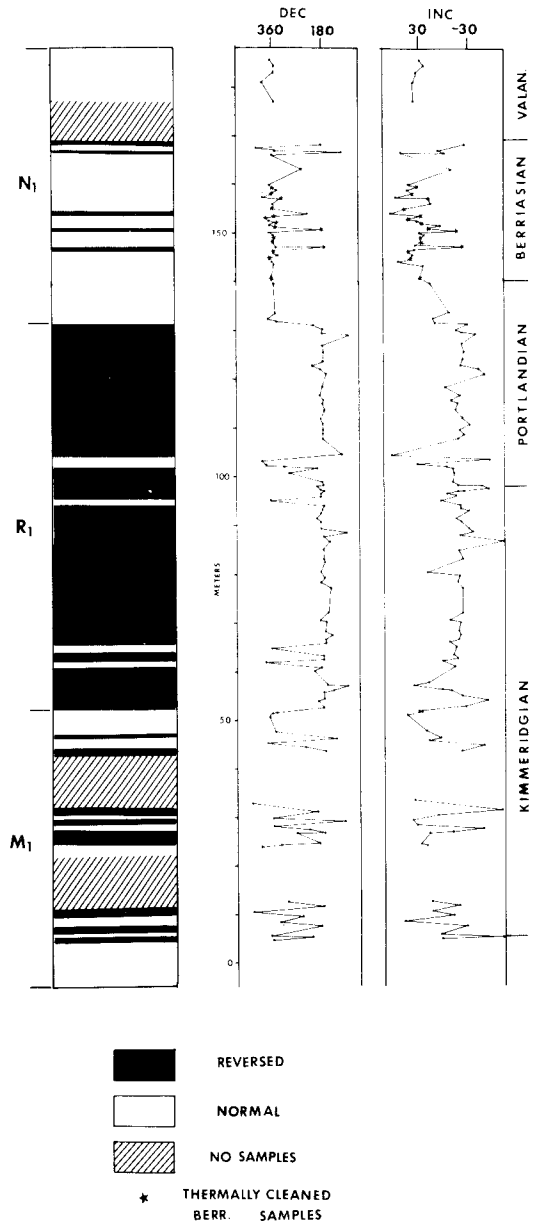


Figure 8. Stratigraphic summary of the declination and inclination values from samples of Kimmeridgian to Lower Valanginian age collected from the slopes of Djebel Oust, with a polarity synthesis. Obliquely lined areas indicate shale regions from which no samples were obtained.

THE CRETACEOUS SECTION (EXCLUDING THE BERRIASIAN)

The lowest Valanginian follows the Berriasian without break on Djebel Oust, the boundary between the two lying close to the change in slope which marks the beginning of the road cut. Their outcrop, however, is not extensive, soon disappearing below the surface cover. The remainder of the Lower Cretaceous section was sampled due east of Djebel Oust. (Memmi & Busnardo 1972), and east of the Zaghouan road in a stream section. The succession

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Table 1. Summary of data from the Upper Jurassic rocks of Djebel Oust, Tunisia. All data used.

		D		I	N	α_{95}	K	Palaeomagnetic pole	
Portlandian limestones	N	354.9	+	18.0	9	16.1	11.1	62.4°S	21.1°E
	R	181.7	–	23.0	32	6.3	17.1	65.3°S	6.2°E
Kimmeridgian limestones	N	347.8	+	25.7	23	19.9	3.2	64.5°S	38.7°E
	R	183.3	–	21.2	49	7.2	8.8	64.2°S	2.7°E
Oxfordian and Callovian	N	346.3	+	21.8	35	4.8	25.4	62.0°S	39.7°E
(in part) ammonitico rosso	R	180.0	–	28.2	7	21.2	9.0	68.6°S	10.1°E
				Mean palaeomagnetic pole				65.2°S	20.3°E $\alpha_{95} = 6.1$

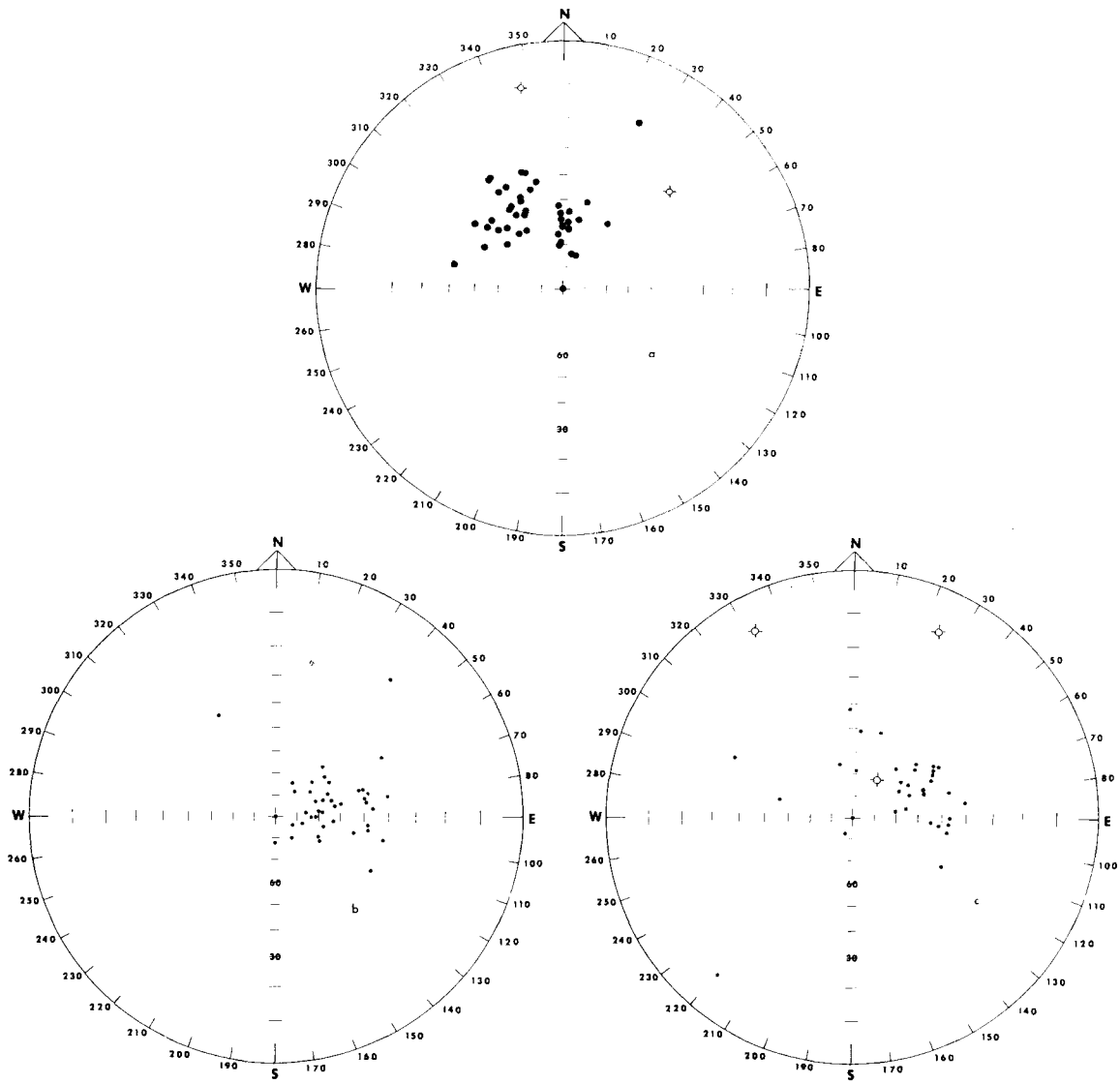


Figure 9. Stereograms of the declination and inclination of Barremian samples: (a) NRM field position; (b) NRM palaeoposition; (c) cleaned directions (500 Oe or 400°C) palaeoposition. Solid symbols indicate projection on the lower hemisphere.

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was complete except that as a result of faulting the Aptian was missing. The Upper Cretaceous section on Djebel Ben Salah extends from the Cenomanian up to the Maestrichtian. The Maestrichtian section is incomplete and is overlain by Eocene rocks. The Cretaceous section is predominantly a limestone shale alternation, usually of thin shales and limestone, but with a thick development of shale above the basal Santonian, in the Upper Maestrichtian and through most of the Valanginian. There is a dominantly carbonate Coniacian and Campanian section. Except for a few thin Valanginian and Barremian sandstone bands considered by Tunisian geologists to be flyschoid in character, arenaceous material is absent.

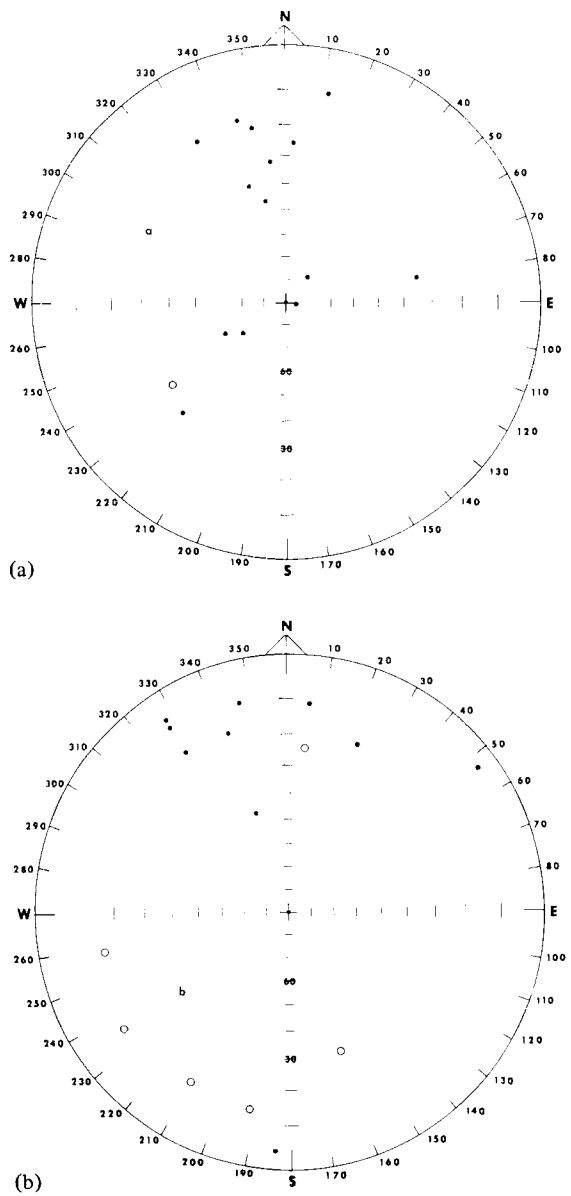


Figure 10. Stereograms of the declination and inclination of Maestrichtian samples; (a) NRM, (b) directions of magnetization cleaned at 450°C. Both figures illustrate directions corrected for the dip of the beds.

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Unlike the sampling of the Jurassic, most of the samples were obtained from surface outcrop, the exception being shallow quarries in the Coniacian chalk and the Lower Santonian limestones, and the Maestrichtian collected in a dry stream bed. The intensity of magnetization of these limestones was appreciably lower than that of the Jurassic samples, most commonly being below 1×10^{-6} emu cm^{-3} so that repeat measurements were necessary when measuring on the Digico. With very few exceptions the application of magnetic cleaning did improve the data scatter, and as the least dispersion occurs with respect to the present horizontal rather than the bedding a present field component appears to be dominant. The results from the measurements of Barremian samples are illustrated by way of example (Fig. 9). The only samples which appeared to yield results were the dry gully and quarry samples and Fig. 10 depicts the NRM and cleaned (450°C) directions of magnetization of Maestrichtian samples from this gully, recorded with respect to bedding.

Discussion

STABILITY OF REMANENCE

Although the Upper Mesozoic section was sampled from Upper Callovian to Maestrichtian with very few breaks in the sequence there is considerable variability in the nature of the results. In a general way, a single stable component of magnetization can be isolated in the Jurassic samples, but only in rare cases can this be done in Cretaceous rocks.

The stable remanent component in the Jurassic rocks is interpreted as primary, i.e. depositional or very early diagenetic. There is a reduction in dispersion when the correction for bedding is made and the rocks respond to thermal and alternating field treatment to give normal and reversed magnetizations which are near antipodal. The remanence in the Kimmeridgian and Portlandian appears to be due to finely divided ferromagnetic material although this has not been observed microscopically. The limestones contain unaltered authigenic pyrite suggesting no oxidization has occurred since the formation of the pyrite. On the basis of the demagnetization experiments, the magnetization of the ammonitico rosso also appears to be due to a finely disseminated magnetite rather than to the more obvious haematite. Although there is still considerable debate concerning the origin of the haematite, the concensus of geological opinion is that the pigmenting material is of primary origin. Some haematite may be disseminated within the limestones with a distribution which does not seem to be controlled by cracks or fissures. When the limestones are nodular it forms a coating on the nodules, as it coats ammonites.

The intensity of magnetization of the Cretaceous surface samples is in general an order of magnitude lower than that of the Jurassic rocks. These rocks do not show a consistent response to magnetic cleaning, and show least dispersion in the field position. We therefore conclude that we are unable to isolate a primary remanence. In thin section pyrite is found pseudomorphed or replaced by limonite and occasionally haematite, the result of supergene alteration. This alteration is, we believe, the source of the secondary magnetization. While this weathering penetrates to a depth greater than that represented by our surface samples, it has not affected the Jurassic rocks from as little as 1 m below the surface. This thin zone of weathering we attribute to the low rainfall and rapid run-off characteristic of the climatic and physiographic conditions. Consistent with this interpretation are the Maestrichtian results from the dry stream bed on Djebel Ben Salah (Fig. 10), where the erosion of periodic streams keeps the rock surface fresh.

Many Cretaceous samples show intermediate directions suggesting the coexistence of oppositely directed components which are equally insensitive to the cleaning applied. There is a fundamental indeterminacy concerning the nature and age of the two components, but

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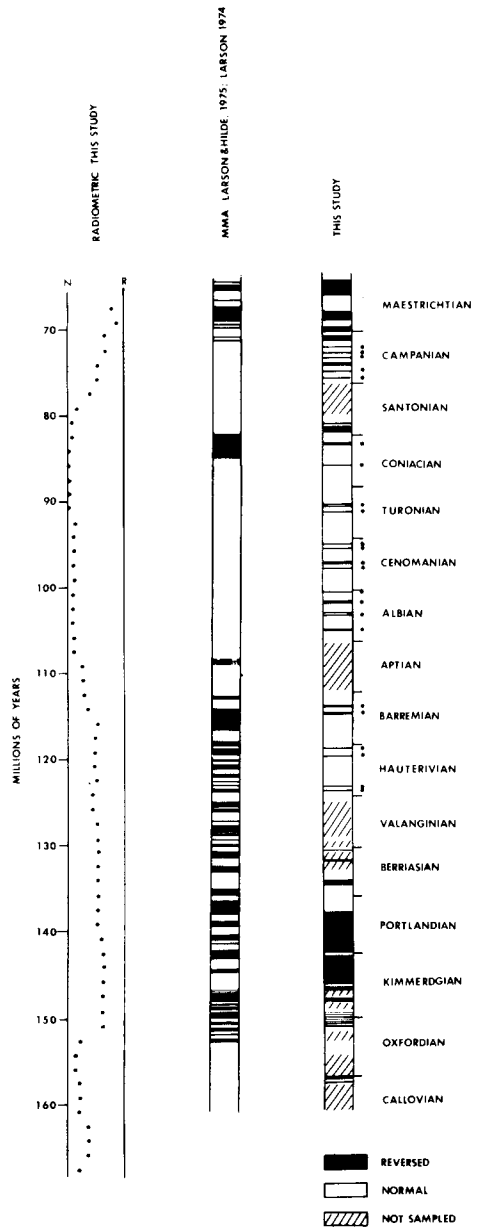


Figure 11. Summary of the stratigraphic position of samples with intermediate directions of magnetization. This figure therefore shows the location of possible Cretaceous polarity changes *if* the interpretation of p. 14 is followed. Their uncertainty is indicated by the use of asterisks.

as one possible interpretation implies the existence of polarity changes in the Cretaceous normal period, the levels at which they occur is noted in Fig. 11.

DIRECTIONS OF MAGNETIZATION

The directions of magnetization recorded for the upper part of the Jurassic are shown in Figs 3 and 7, and summarized in Table 1, where means have been calculated for both normal

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and reversely magnetized rocks. All data have been used, for use of the empirical definition of anomalous directions (Wilson, Dagley & McCormack 1972) would lead to the exclusion of less than ten measurements. While the reversed and normal directions are not antipodal, the circles of confidence do overlap so that the difference is not significant.

The mean palaeomagnetic pole for the Upper Jurassic, obtained from the above data 65.2°S 20.3°E $\alpha_{95} = 6.1$, lies fairly close to that obtained from the Moroccan Liassic turbidites and is within the grouping of African Jurassic poles.

THE POLARITY RECORD

Based upon the approximate calculation on the rates of limestone deposition and assuming that limestone and shale depositional rates are approximately equal, perhaps an unwarranted assumption, a sampling interval of 25 to 45 cm should permit the recovery of a complete polarity record where the frequency of polarity changes is no greater than during the Cenozoic. This, however, does not apply to a condensed sequence such as the ammonitico rosso. With the exception of the thicker shale intervals, the sampling of the Upper Jurassic Kimmeridgian and Portlandian was of the necessary density although there could be as little as one sample in a polarity zone of short duration.

The record from the ammonitico rosso is incomplete for in the two principal sections there were two principal shale intervals from which no samples could be obtained and within which polarity change could easily occur.

For reasons indicated earlier (p. 13) no conclusions can be drawn from most of the Cretaceous data below the Maestrichtian. The strongest suggestion is of a Lower Santonian reversal from samples collected in a shallow quarry which occurs at about the horizon reported by Perchersky (Creer 1971) and a less well-defined zone in the Lower Campanian. The Lower Maestrichtian samples (Fig. 8) together with our topmost Campanian give a sequence similar to Lowrie & Alvarez (1977a, b) and Alvarez *et al.* (1977), without their two short duration reversals.

MARINE MAGNETIC ANOMALIES

The third source of polarity information lies in the interpretation of marine magnetic anomalies. The record is not complete for, as a consequence of the distance of the towed magnetometers from the anomaly source, field reversals of short duration will not always be recorded. This is particularly true if the source of the anomaly, or an important contribution to it, lies deeper in the crust than is customarily thought (Kent *et al.* 1978).

Some time ago, Baldwin, Coney & Dickinson (1974) discussed the indeterminacy of the Cretaceous marine magnetic anomaly scale, with particular reference to the Cretaceous quiet (normal) period. This indeterminacy was resolved by the analysis of Irving & Pullaiah (1976) and confirmed by the version of the analysis used here. Subsequently a more complete model for the polarity history was presented by Larson & Hilde (1975), and palaeomagnetic comparisons made by Lowrie & Alvarez (1977a, b) who have the most complete Cretaceous data, by Keating, Helsley & Pessagno (1975) and by Hanna (1973). Because of the instability of most of our Cretaceous data we cannot add significantly to the discussion of the Cretaceous. The Jurassic data, however, we regard as reliable and may be applied to consideration of the Keathley anomalies. Although the radiometric data are unfortunately few, there appears to be a change from predominantly normal magnetization to mixed polarities at about 152 Myr. The Keathley sequence begins about 153 Myr, that is about the Oxfordian–Kimmeridgian boundary, the first marine magnetic anomalies from M 25 to M 22 (Larson & Hilde 1975) appearing as a low amplitude pulse increasing to characteristic amplitudes by

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M 22. After M 22 reversals are frequent with little bias towards either normal or reversed directions up to the Cretaceous quiet period.

If we take the change in field behaviour at the base of the Keathley sequence as a key, the best parallel is with the zone of frequent reversals in the Upper Oxfordian Lower Kimmeridgian. It would suggest that the assignment of 153 Myr to the Oxfordian–Kimmeridgian boundary is too old by a few million years (see Fig. 11). It would also follow that the remainder of the Callovian must belong to the Jurassic quiet period. However, it must be noted that in our data as well as in that of Steiner & Helsley (1975), Creer, Miller & Smith (1965) and that of Pechersky (Creer 1971), reversed polarity does occur within that period.

The marine magnetic anomaly data also implies that within the Lower Cretaceous more reversed horizons remain to be discovered in terrestrial sequences and, if correct, implies that the apparent bias to normal and then reversed directions in the palaeomagnetic record may be spurious.

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